Ethiopian Association of Anesthetists

Evidence Based Clinical Decision Making

Participants' Manual

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Abbreviation

EAA	Ethiopian Association of Anesthetists
AHA	American Heart Association
EBCDM	Evidence Based clinical Decision Making
EBDM	Evidence Based Decision Making
EBM	Evidence Based Medicine
EBP	Evidence Based Practice
GRADE	Grading of Recommendations Assessment, Development and Evaluation
QI	Quality Indicators
RCT	Randomized Controlled Trial
USPSTF	US Preventive Services Task Force

Introduction to the Manual

This Evidence Based decision making (EBDM) training is 5 days training designed to prepare health professionals to critically appraise and implement available evidences for clinical decision making.

Evidence-based medicine (EBM) is a conscientious, explicit, and judicious use of current best evidence in making decisions about the care of individual patients. The idea has now been expanded to encompass the relationship between patients and healthcare professionals in clinical practice, as well as the integration of patient preferences with the health care professionals experience and the best available research evidence in the decision-making process.

Generally, EBDM practice can be applied in five steps: step one; converting the current clinical scenario into an answerable question, step two; identifying the best available research evidence by performing a proper literature review across various databases, step three; critical appraisal of the evidence for its validity, impact, and applicability, step four; applying the results of the appraisal with the clinical experience and patient's values, and step five; evaluating the process and finding ways to improve it in the next time.

Applying evidence based descion making will improve diagnosis, clinical judgment, decision making and patient outcomes. Despite its utmost importance in developing countries for its cost-effectiveness and efficient use of healthcare resources, training need assessments has identified the need for the course for health professionals. Moreover, because of the importance of EBDM in building clinical practice on a scientific basis, there is a focus and a need on providing continuous training and capacity building programmes for health professionals. Evidentiary based skills and information accessible to clinicians by developing reliable clinical manual is also paramount.

The aim of this evidence based decision making course participant manual is to provide guidance, additional information, tools and resources to improve their knowledge, skill and attitude in providing evidence based clinical care. The training manual will also serve as a reference in all phases of the training. The manual is organized into five (5) chapters to address the most relevant themes in evidence based medicine and addresses gaps that were identified during the training needs assessment. Furthermore, each steps of evidence based medicine practice steps are include in detailed to improve participants' skills, knowledge and behaviors in integrating evidences into their decision making for individual patients.

Health care professional's failure to engage in EBDM has repercussions for the health of individuals and populations and it is strongly believed that the training will improve the evidence based clinical practice related gap. In addition, it improves both patient outcomes and trainees' practice-based learning and improvement. Different teaching modalities like interactive lectures, computer laboratory and clinical scenario based teachings and discussions will be used.

Rationale

EBDM is an approach that prompts professionals to use quantitative and qualitative data to develop informed plans. Evidence based decision making is essential for health professionals in order to provide the best care for their patients. It is important for health professionals to be aware of the latest research and best practices in order to make informed decisions that are tailored to the individual needs of patients.

Training on EBDM can help health professionals stay up-to-date on the latest research, understand the implications of different treatments, and make decisions that are in the best interest of their patients. EBDM training can also help to reduce medical errors, improve patient safety, and ensure the best outcomes for patients. With the right training, health professionals can ensure that their decisions are based on the most current, accurate, and reliable evidence.

Training health professionals in evidence-based decision-making can also help ensure that patients receive the best possible care. A training need assessment (TNA) conducted by EAA indicated that there is a gap among professionals in using Evidence based clinical decision making. Therefore, this training manual is prepared to encourage health professionals to apply evidence-based decision making in their clinical practice.

Target Audiences

This EBDM in-service training module is designed to equip all **health professionals** in the knowledge, skills and attitudes necessary to apply evidence based clinical decision-making practices.

Course syllabus

1. Module name: Evidence based Decision making

2. Module code:

3. Module duration: 5 days

4. Prerequisite: Health professionals with a minimum of BSc degree in their respective discipline

5. Module continuing education unit (CEU):

6. Module description: This four days training module/course is designed for all health professionals to equip them with all the essential knowledge, skills and attitudes on formulating relevant clinical questions, identifying hierarchy of evidence, conducting evidence retrieval, performing critical appraisal and incorporating up to date evidence in to clinical practice. The training will enable trainees make more informed decisions supported by the most current and up to date information which lead to better patient outcomes.

7. Module/course goal: at the end of this training, trainees will be able to make a sound clinical decision based on principles of evidence-based medicine.

8. Enabling/learning objectives

- Formulate clinically answerable questions
- Identify appropriate sources of evidence
- Critically appraise selected evidences to determine its validity, relevance, and applicability
- Implement EBDM principles in everyday clinical practice
- Evaluate the outcomes of interventions and clinical decisions

9. Chapters/sessions outline

Table 1: Chapter/session outline

Chapter	Title /topic	Durations	Sessions
1	Introduction for EBM and Formulating Answerable Clinical Questions	**5 Hrs**	• Introduction to Evidence Based Clinical Decision Making (EBCDM)(2 Hr) ○ Definition ○ Benefits of EBM ○ Standards and criterias for EBM • Formulating an Answerable Clinical Question (3Hrs) ○ Introduction ○ How to ask well formulated answerable questions? ○ The type of question being asked ○ The type of study that best answers the question
2	Evidence retrieval	**13 Hrs**	• Study Design and Strength of Evidence for Relevant Clinical Question (3 Hrs) • Sources of evidence (4 hrs) ○ The Haynes 5S knowledge acquisition pyramid ○ Study designs and Strength of Evidence • Effective Search Strategy (2 hrs) • Effective Database search (4 Hrs)
3	Appraising the evidence	**11 hour**	• Critical Appraisal (4Hrs) • Critical Appraisal Checklist (7)
4	Applying Evidence to Practice	**5 Hrs**	• Clinical Pathways and Decision analysis (3Hrs) ○ Clinical Pathways ○ Practice Guidelines ○ Decision Analysis • Assessing Recommendations and Facilitating Implementation of Evidence Based Practice (2Hrs) ○ Assessing Recommendations ○ Fascilitator and Barriers to the Implementation of Evidence into Practice
5	Assess Outcomes, Make Changes to Practice, and Quality Assurance	**4 Hrs**	Assess Outcomes and Make Changes to Practice (1 Hr) ○ Introduction ○ How to measure changes ○ Disseminating the results of an EBP intervention ○ How to assess your own EBP performance • Quality Assurance (1 Hr) ○ Quality indicators ○ Quality indicator assessment

10. Teaching learning methods and activities

- Interactive Lecture
- Group work
- Case studies/presentation
- Clinical scenario/group work/presentation
- Journal club critical appraisal/individual work/presentation
- Database searching presentation
- Role play and simulation (e.g. communicating evidence to patient)
- Hands on exercise (e.g. writing clinical questions)

11. Learning materials and resource (including references)

- Training manual
- Scientific journals and articles
- EBDM text books
- Access to research databases like Cochrane, HINARI or Pubmed database
- Prepared case studies/case scenarios
- slides displaying research study flow, baseline data, and results of the EBDM topic in discussion
- videos or webinars (application of EBDM on real world practice)
- Critical appraisal guidelines
- Clinical guidelines
- EBDM check list for implementation
- Writing boards
- Posters/pictures
- Laptops
- White board, marker
- LCD projector
- PPT

12. Reference

- *David L. Sackett, Sharon E.. Straus, Richardson WS. Evidence-based medicine: how to practice and teach EBM. Churchill Livingstone; 2000.*

- *Prasad K. Fundamentals of evidence based medicine. New York: Springer; 2013.*
- *Gøtzsche P. Rational diagnosis and treatment: Evidence-based clinical decision-making. John Wiley & Sons; 2008 Mar 11.*
- *Fleisher LA. Evidence-Based Practice of Anesthesiology E-Book. Elsevier Health Sciences; 2013 Jan 15.*
- *Hecht L, Buhse S, Meyer G. Effectiveness of training in evidence-based medicine skills for healthcare professionals: a systematic review. BMC medical education. 2016 Dec;16(1):1-6.*
- *Zwolsman SE, van Dijk N, Te Pas E, Wieringa-de Waard M. Barriers to the use of evidence-based medicine: knowledge and skills, attitude, and external factors. Perspectives on medical education. 2013 Feb;2(1):4-13.*
- *Shaneyfelt T, Baum KD, Bell D, Feldstein D, Houston TK, Kaatz S, Whelan C, Green M. Instruments for evaluating education in evidence-based practice: a systematic review. Jama. 2006 Sep 6;296(9):1116-27.*
- *Shafaghat T, Bastani P, Nasab MH, Bahrami MA, Montazer MR, Zarchi MK, Edirippulige S. A framework of evidence-based decision-making in health system management: a best-fit framework synthesis. Archives of Public Health. 2022 Dec;80 (1):1-20.*
- *Greenhalgh T. How to read a paper: The basics of evidence-based medicine. John Wiley & Sons; 2014 Apr 7.*

13. Learning Assessment methods

- Pre & posttest
- Participant feed back
- Quiz, exercises

14. Trainer and facilitators selection criteria

- Facilitators will be those who have developed this manual and those who undertaken the trainers course and have TOT on evidence based decision making

15. Trainees and participants selection criteria

- Health care professional providing clinical service in different units.

16. Training evaluation methods (satisfaction, learning, behavior and impact)

- Surveys
- Interviews
- Observation
- Pre and posttest

17. Venue - All CPD accredited institutions

18. Composition

19. Schedule

Table 2: Training Schedule

Time (LT)	Day 1	Day 2	Day 3	Day 4	Day 5
2:30-3:30	○ Welcome and introduction ○ Participant registration ○ Self-introduction and expectation ○ Pretest (MCQ)	**Warm up and agenda** **Cont.** **Evidence retrieval** ○ Study Design and Strength of Evidence for Relevant Clinical Question ○ Study design types ***Activities***	**Warm up and agenda Cont.** **Evidence retrieval** **Sources of evidence** ○ The knowledge acquisition pyramid ○ filtered and unfiltered sources of evidence ***Activities***	**Warm up and agenda Cont.** **Appraising the Evidence** ○ Critical Appraisal Checklist ***Activities***	**Warm up and agenda Cont.** **Applying Evidences to Practice** ○ Clinical Pathways and Decision analysis ***Activities***
3:30-4:30	**Introduction to Evidence Based Clinical Decision Making** • Definition • Benefits of EBM • Standards and criterias for EB ***Activities***		**Evidence retrieval** ○ Effective Database search ***Activities***	**Appraising the Evidence** ○ Critical Appraisal Checklist ***Activities***	**Applying Evidences to Practice** ○ Clinical Pathways and Decision analysis ***Activities***
4:45-6:30	**Introduction to Evidence Based Clinical Decision Making** ○ Standards and	**Evidence retrieval** ○ Study Design and Strength of Evidence	**Evidence retrieval** ○ Effective Database search	**Appraising the Evidence** ○ Critical Appraisal Checklist	**Applying Evidences to Practice** ○ Assessing Recommendatio

	criterias for EBM Formulating an Answerable Clinical Question ○ Introduction ○ How to ask well formulated answerable questions? ***Activities***	for Relevant Clinical Question ○ Identifying types of study design for clinical question ***Activities***	***Activities***	***Activities***	ns and Facilitating Implementation of EBP ***Activities***
LUNCH	Lunch break				
8:00-9:30	**Formulating an Answerable Clinical Question** ○ The type of question being asked ○ The type of study that best answers the question ***Activities***	**Evidence retrieval** **Effective Search Strategy** ○ ***Activities***	**Appraising the Evidence** ○ Critical Appraisal ***Activities***	**Appraising the Evidence** ○ Critical Appraisal Checklist ***Activities***	Assess Outcomes, Make Changes to Practice, and Quality Assurance ***Activities***
9:00-9:45	**Formulating an Answerable Clinical Question** ○ The type of question being asked ○ The type of study that best answers the question ***Activities***	**Sources of evidence** ○ The knowledge acquisition pyramid ○ filtered and unfiltered sources of evidence ***Activities***	**Appraising the Evidence** ○ Critical Appraisal ***Activities***	**Appraising the Evidence** ○ Critical Appraisal Checklist ***Activities***	Post test
9:45-11:00			**Appraising the Evidence** ○ Critical Appraisal ***Activities***	**Appraising the Evidence** ○ Critical Appraisal Checklist***Activities***	**Feed back and closing**

Chapter One: Introduction for EBM and Formulating Answerable Clinical Questions

Duration: 4 hours

Primary Objective

The objective of this chapter is to provide background information about evidence based clinical decision making and enable trainees to formulate relevant answerable clinical questions.

Enabling Objective

At the end of this session the trainee will be able to

- Define evidence based clinical decision making (EBCDM)
- Explain the benefits of EBCDM
- Describe the 5 steps of EBCDM / EBM
- Categorize evidences in to different levels based on their strength of evidence
- Formulate an answerable relevant clinical question using the PICO (T) framework
- Differentiate the type of question being asked
- Identify the type of study which best answers the formulated question

Session1.1: Introduction to Evidence Based Clinical Decision Making (EBCDM)

Session Duration: 1 hour

Session Objective: At the end of this session, trainees will be able to:

- Define evidence based clinical decision making (EBCDM)
- Explain the benefits of EBCDM
- Describe the 5 steps of EBCDM / EBM
- Categorize evidences in to different levels based on their strength of evidence

Summary of Activity for Session 1: Introduction to EBM

No	Activity	Method of delivery	Time	Slide Number	Resource
1.	Think Pair Share	• Group discussion • Reflection	20 mins		LCD Flip Chart Computer
2.	Definition of EBM	• Interactive Lecture	5 mins		

3.	Benefits of EBM	• Interactive lecture • Group Discussions	7 mins		markers
4.	Standards and criteria for EBM	• Interactive lecture • Individual reflection	25 mins		
5.	Summary	• Interactive presentation	3 min		

Learning activities for Session 1: Introduction to EBM

Time	Methods and activities	Material and resource
60 minutes	**Think pair share (20mins)**: - The facilitator will introduce the session objectives and ask the trainees what EBM is and its benefits. Allow them to think, pair in to small groups and then share their answers to the whole group **Interactive presentation: 37 mins** Definition, benefits, standards and criteria for EBM **Summarize the session (3 mins)**: The facilitator summarizes showing key points of the session	LCD Flip Chart Computer markers Reference manual

Summary of Activity for session 2: Formulating an answerable clinical question

No	Activity	Method of delivery	Time		Resource
1	Think Pair Share	• Group discussion • Reflection	20 mins		LCD Flip Chart Computer Markers Case scenarios
2	Session introduction	• Interactive Lecture	5 mins		
3	How to ask well formulated, answerable questions	• Interactive lecture • Reflection	15 mins		
4	Group discussion based on case study	• Small group discussion	45 mins		

5	The type of question being asked	• Interactive lecture • Reflection	15 mins		
	The type of study that best answers the question	• Interactive lecture • Reflection	15 mins		
	Group discussion based on case study	• Small group discussion	20 mins		
	Group discussion based on case study	• Small group discussion	40 mins		
6.	Summary	• Interactive presentation	5 mins		

Learning Activity for Session 2: Formulating an answerable clinical question

Time	Methods and activities	Material and resource
180 minutes	**Think pair share (20mins)**: - The facilitator will introduce the session objectives and ask the trainees about the importance of formulating a good question and what question formulating formats they know. Allow them to think, pair in to small groups and then share their answers to the whole group **Session introduction (5 mins)** The facilitator will give brief presentation about the importance of formulating good clinical question as the 1st step of EBM **Interactive presentation: 15 mins** The facilitator will give an interactive presentation about how to ask well formulated, answerable questions using a PICO format **Group discussion (45 mins)** ***Introduce the case*** *(5* ***min)*** - Read/ display the case scenario - The facilitator organizes the participants in to small groups with 4/5 members in each group **- Tell them to read the case study – 5 mins**	LCD and Computer Flip chart and marker Case scenarios Reference manual

	Allow the group to analyze the case (15 min): **Give the opportunity to react to the whole group (15 min**): **Summarize the cases (10 min):** The facilitator summarizes showing key concepts of the case study **Interactive presentation: 30 mins** The facilitator will give an interactive presentation about the type of question being asked and the type of study that best answers the question **Group discussion (20 mins)** - The facilitator organizes the participants in to small groups with 4/5 members in each group **-** give the opportunity to analyze the case and questions (10 mins) **- allow them to reflect to the whole group (5 mins)** **Summarize the cases (5 min):** The facilitator summarizes showing key concepts of the case study **Group discussion (40 mins)** ***Introduce the case*** *(5* ***min)*** - The facilitator organizes the participants in to small groups with 4/5 members in each group -distribute the printout case scenario to groups (different scenarios to each group) **-** Tell them to read the case study – 5 mins ***Allow the group to analyze the case (15 min):*** **Give the opportunity to react to the whole group (15 min**): **Summarize the cases (5 min):**	

	The facilitator summarizes showing key concepts of the case study **Summarize the session (5 mins)**: The facilitator summarizes the session showing key points of the session	

Chapter Two: Evidence retrieval

Duration - 11 hour

Enabling objectives

- Identify study designs for relevant clinical questions
- Formulate an effective search strategy for evidence
- Describe sources of evidence
- Perform efficient database search

Session 2.1: study design and strength of evidence for relevant clinical question

Duration: 3 Hrs

Session objectives: by the end of this session trainees will be able to

- Types of study design
- Identify types of study design for clinical question

Summary of Activity: study design and strength of evidence for relevant clinical question

No	Activity	Method of delivery	Time	Slide Number	Resource
7.	Study design and strength of evidence	• Think Pair Share • Case scenario • Interactive Lecture • Reflection activities	140 min		LCD Flip Chart Computer markers
8.	Identify types of study design for clinical question	• Interactive lecture	30min		
9.	Summary	• Present the final session • Ask questions and comments.	10 min		

Learning activities: study design and strength of evidence for relevant clinical question

	Topic	Time	Activity
1	Study design and strength of evidence	140 min	Think pair share What kinds of study designs are available for clinical questions? What are their pros and cons? **Duration:** 10 min **Instruction** • T: (Think): "think" about what you know about the above questions and write your answer - 5 minutes • P: (Pair): pair with other participants or a small group (4 or 5 per group) and discuss your answers – 5 minutes • S: (Share): share your thinking with the whole group-10 minutes **Lecture 90 min** • Discuss on clinical question and study design • Discuss individual study designs and their strength • Discuss descriptive studies • Discuss longitudinal studies • Discuss qualitative study designs • Discuss systematic review and meta-analysis • Discuss on quality of evidence of comparative studies **Case scenario 20 min** Identify the possible study design for the following clinical questions • What is the incidence of nausea and vomiting after tympanoplasty surgery? • What is the most effective treatment for increased intraocular pressure?

			• What is the impact of obesity on pregnancy related complications? • What is the association between caffeine consumption and the incidence of atrial fibrillation? **Instruction** • Make a group of 4/5 members • Discuss on the scenario within groups for 15 minutes Share your answer to the whole group – 15 minutes **Reflection activities** **Duration – 20** mins **Instruction and question:** Be in group of 2 and develop clinical questions for the following study designs • Case study • Cross sectional research • Case control • Cohort • Qualitative study design
2	Identify types of study design for clinical question	30min	Lecture • Discuss strength of comparative studies • Discuss relationship of clinical question to study design
3	Summary	10min	• Present summary • Ask questions and take comments

Session 2.2: Effective Search Strategy

Duration: 3 Hrs

Session objectives

By the end of this session trainees will be able to

- Design search strategy
- Explain the methods of searching for evidence

Summary of Activity: effective search strategy

No	Activity	Method of delivery	Time	Slide Number	Resource
1.	Steps in Search for evidence	• Think Pair Share • Interactive Lecture	110 min		LCD Flip Chart Computer markers
2.	Summary	• Present the final session • Ask questions and comments.	10 min		

Learning activities: effective search strategy

	Topic	Time	Activity
1	Steps in Search for evidence	110 min	**Think-Pair-Share** • How do you search for reliable source of information? • What key words can be used for searching for a randomized control trial of the effectiveness of lidocaine for alleviating post operative sore throat (therapy study)? • What are the best ways to search for evidence on HIV/AIDS **Duration** – 30 min **Instruction**

			❖ T: (Think): "think" about what you know about the above 2 questions and write your answer - 5 minutes ❖ P: (Pair): pair with other participants or a small group (4 or 5 per group) and discuss your answers – 5 minutes • S: (Share): share your thinking with the whole group-10 minutes **Lecture 80 min** • Discuss steps of searching evidence • Explain the use of MeSH heading
2.	Summary	10 min	• Present the final session • Ask questions and comments.

Session 2.3: Sources of evidences

Duration:

Session objectives

By the end of this session trainees will be able to

- Explain the knowledge acquisition pyramid
- Describe filtered and unfiltered sources of evidence

Summary of Activity: Sources of evidence

No	Activity	Method of delivery	Time	Slide Number	Resource
1.	The Haynes 5s knowledge acquisition pyramid	• Group discussion • Interactive Lecture	110 min		LCD Flip Chart Computer markers
2.	Describe filtered and unfiltered sources of evidence	• Interactive lecture • Reflection activities	120 min		
3.	Summary	• Present the final session • Ask questions and comments.	10 min		

Learning activities: sources of evidence

	Topic	Time	Activity
1	The Haynes 5s knowledge acquisition pyramid	110 min	**Group discussion** 1. What is the knowledge acquisition pyramid and how can it help us understand EBM? 2. What are the different levels of knowledge in EBM and how can they be used to improve our practice? **Duration – 20** mins **Instruction** • Make a group of 5 members • Discuss with groups for 15 minutes • Share your answer to the whole group – 15 minutes **Lecture 90 min** • The Haynes 5s knowledge acquisition pyramid • Discuss primary, secondary and tertiary literature
2	Describe filtered and unfiltered sources of evidence	120 min	**Activity** - pair up and search for evidence • Does a low-carbohydrate diet result in better glycemic control compared to a conventional diet in pregnant women with gestational diabetes, • Is physical therapy result better pain relief compared to medication alone for patients with chronic back pain? • In patients with type 2 diabetes, does a diet and exercise program result in better glycemic control compared to a medication-only approach? • What is the best treatment option for myalgia? **Duration: 20 minutes**

			Instruction: Pair up and search evidence for the above topics **Lecture 100 min** scuss filtered and unfiltered sources of evidence
3.	Summary	10min	• Present the final session • Ask questions and comments.

Session 2.4: Effective database search

Session objectives: By the end of this session trainees will be able to **search** for evidence efficiently

Summary of Activity: effective database search

No	Activity	Method of delivery	Time	Slide Number	Resource
1.	Database search	• Think Pair Share • Interactive Lecture	230 min		LCD Flip Chart Computer markers
2.	Summary	• Present the final session • Ask questions and comments.	10 min		

Learning activities: effective database search

	Topic	Time	Activity
1	Database search	230 min	Think-pair-share 30min • What keywords should I use to find the best sources of evidence on current COVID treatment • What search terms should I use to find evidence on malaria therapy? Duration: 20 min **Instruction**

			• T: (Think): think about what you know about the above questions and write your answer - 5 minutes • P: (Pair): pair with other participants or a small group (4 or 5 per group) and discuss your answers – 5 minutes • S: (Share): share your thinking with the whole group-10 minutes **Lecture 180 min** • Discuss searching tips • Discuss searching with mesh • Discuss searching without mesh Activity **Think pair share 20 min** Convert the following clinical questions in to well-built search strategies • *In patients with osteoarthritis of the knee is hydrotherapy more effective than traditional physiotherapy in relieving pain?* • Does acupuncture help reduce pain and improve function in people with rheumatoid arthritis? • Is cognitive behavioral therapy more effective than traditional pharmacological treatment for treating post-traumatic stress disorder? • What is the most effective treatment for reducing symptoms of anxiety in children? • Is there a difference between the effectiveness of antibiotics and antiviral medications in treating acute bronchitis?
2.	Summary	10 min	• Present the final session • Ask questions and comments.

Chapter Three: Appraising the Evidence

Duration: 9 hours

Primary Objective

- The objective of this chapter is to enable trainees to critically appraise evidence and apply it in clinical setting.

Enabling Objective

At the end of this session the trainee will be able to

- Outline the concepts of critical appraisal.
- Use methods for identifying valid evidence.
- Apply critical appraisal tools for therapy, prognosis, and diagnosis

3.1 Summary of Activity for session 1: Critical appraisal

No	Activity	Method of delivery	Time		Resource
10.	Think Pair Share	• Group discussion • Reflection	30 mins		LCD Flip Chart Computer markers
11.	Introduction	• Interactive Lecture	10 mins		
12.	What is the PICO of the study, and is it close enough to your PICO?	• Interactive lecture • Individual reflection	30 mins		
13.	How well - or how badly - was the study conducted?	• Interactive lecture • Individual reflection	40 mins		
14.	What are the results and are they applicable to my patients and my clinical setting?	• Interactive lecture • Individual reflection	30 mins		
15.	Summary	• Interactive presentation	10 mins		

Learning activities for session 1: Critical appraisal

Time	Methods and activities	Material and resource
60 minutes	**Think pair share (30 mins)**: - The facilitator will introduce the session objectives and ask the trainees what critical appraisal is and make them discuss the major components of critical appraisal. Allow them to think, pair in to small groups and then share their answers to the whole group **Interactive presentation: 120 mins** **Discuss the components of critical appraisal in detail** - Start with an introduction to the session (10 mins) - What is the PICO of the study, and is it close enough to your PICO? (30 mins) - How well - or how badly - was the study conducted? (40 mins) - What are the results and are they applicable to my patients and my clinical setting? (30 Mins) **Summarize the session (10 mins)**: The facilitator summarizes showing key points of the session	LCD Flip Chart Computer markers Reference manual

3.2 Summary of Activity for session 2: Critical appraisal Checklist

No	Activity	Method of delivery	Time		Resource
1	Session introduction	• Interactive Lecture	15 mins		LCD Flip Chart Computer Markers
2	Critical appraisal checklist for RCT	• Interactive presentation • Reflection	30 mins		
3	Critical appraisal checklist for a systematic review	• Interactive presentation • Reflection	30 mins		
4	Critical appraisal checklist for cohort, case-control,	• Interactive presentation • Reflection	30 mins		

	cross sectional and qualitative studies				
5	Group discussion: critically appraising evidences	• Small group discussion	4 hours		
6	Summary	• Interactive presentation	15 mins		

Learning Activity for Session 2: Critical appraisal Checklist

Time	Methods and activities	Material and resource
180 minutes	**Session introduction (15 mins)** The facilitator will give brief presentation about what critical appraisal checklist is and explain the importance of these check lists **Interactive presentation: 30 mins** The facilitator will give an interactive presentation about each component of a critical appraisal checklist for RCT. **Interactive presentation: 30 mins** The facilitator will give an interactive presentation about each component of a critical appraisal checklist for RCT. **Interactive presentation: 30 mins** The facilitator briefly introduces about the critical appraisal checklists for cohort, case-control, cross sectional and qualitative studies. **Group discussion (45 mins)** ***Introduce the case*** *(5* ***min)*** - Read/ display the case scenario	LCD and Computer Flip chart and marker Case scenarios Reference manual

	- The facilitator organizes the participants in to small groups with 4/5 members in each group **- Tell them to read the case study – 5 mins** ***Allow the group to analyze the case (15 min):*** **Give the opportunity to react to the whole group (15 min**): **Summarize the cases (10 min):** The facilitator summarizes showing key concepts of the case study **Interactive presentation: 30 mins** The facilitator will give an interactive presentation about the type of question being asked and the type of study that best answers the question **Group discussion (20 mins)** - The facilitator organizes the participants in to small groups with 4/5 members in each group **-** give the opportunity to analyze the case and questions (10 mins) **- allow them to reflect to the whole group (5 mins)** **Summarize the cases (5 min):** The facilitator summarizes showing key concepts of the case study **Group discussion (40 mins)** ***Introduce the case (5)*** - The facilitator organizes the participants in to 4 small groups -distribute the study to be critically appraised to groups (different study type to each group)	

	- Give them to read the study individually – 15 mins ***Allow the group to critically appraise the study (15 min):*** **Give the opportunity to react to the whole group: 120 mins** **(30 min for each group**): **Summarize the appraisal (10 min):** The facilitator summarizes showing key concepts of critical appraisal **Summarize the session (15 mins)**: The facilitator summarizes the session showing key points of the session	

Chapter Four: Applying Evidences to Practice

Duration: 4 hours

Primary Objective: At the end of this module participants will be able to recognize principles of evidence based medicine application to make clinical decisions that are in the best interest of the patients taking into account the potential risks and benefits of any intervention.

Enabling objectives: At the end of this session participants will be able to:

- Define clinical decision analysis
- Discuss concepts of clinical decision analysis and pathway for effective application of evidence into practice
- Identify several models used to guide healthcare professionals in the successful implementation of evidence into practice.
- Describe facilitators and barriers for effective implementation of evidence based practice
- Provide evidence based care for individual patient, taking into account their values, preferences, and clinical context

Session 4.1: Clinical Pathways and Decision analysis

Session Duration: 3 Hrs

Specific Objectives: at the end of this session trainees are expected to

- Describe the clinical pathways and their role in maximizing patient safety and clinical efficiency
- Discuss concepts of clinical Decision analysis
- Perform decision analysis using decision tree approach

Summary of Activity: Clinical Pathways and Decision analysis

No	Activity	Method of delivery	Time	Slide Number	Resource
16.	Definition	• Think Pair Share • Interactive Lecture	10min		LCD Flip Chart

17.	Importance of clinical descion analysis and clinical pathways	• Think Pair Share • Interactive lecture	10 min		Computer markers
18.	Clinical Pathways	• Interactive lecture	90min		
19.	Decision Analysis	• Interactive lecture	60min		
20.	Questions	• Any questions	5min		
21.	Summary	• Present the final session • Ask questions and comments.	5 min		

Learning activities: Clinical Pathways and Decision analysis

Time	Activity	Material and resources
180	**Activity : Think-Pair-Share** **Think pair share (20mins)**: • The facilitator will introduce the session objectives and ask the trainees about clinical descion analysis and the importance of clinical descion analysis and clinical pathways. Allow them to think, pair in to small groups and then share their answers to the whole group **Session introduction (10 mins)** The facilitator will give brief presentation about the session **Interactive presentation: 90 mins** The facilitator will give an interactive presentation about	LCD and Computer Flip chart and marker Case scenarios Reference manual

	• clinical pathways **Interactive presentation: 60 mins** • Decision analysis	

Session 4.2: Assessing Recommendations and Facilitating Implementation of Evidence Based Practice

Session Duration: 3Hrs

Specific Objectives: at the end of this session trainees are expected to

- Describe the importance of assessing recommendation to evidence based clinical decision making
- Identify the barriers and facilitators of evidence based practice implementation
- Assess recommendations of different guidelines using criterias of recommendation assessment

Summary of Activity: Assessing Recommendations and Facilitating Implementation of Evidence Based Practice

No	Activity	Method of delivery	Time	Slide Number	Resource
1.	Assessing Recommendations	• Think Pair Share • Interactive Lecture	20min		LCD Flip Chart Computer markers
2.	Barriers and facilitators of evidence based practice implementation in your institutions?	• Think Pair Share • Interactive lecture	25 min		

Learning activities: Introduction to Evidence Based Clinical Decision Making (EBCDM)

Time	Activity	Material and resources
180 mins	**Think-Pair-Share** The facilitator will introduce the session objectives and ask the trainees about the barriers and facilitators of evidence based practice implementation in their institutions? **Duration – 25min**	LCD and Computer Flip chart and marker Reference manual

	Instruction o T: (Think) about what you know or have learned about the above question- 7minutes o P: (Pair) with another participant or a small group and discuss the points - 8minutes o S: (Share) your thinking with your partner and facilitators expand the "share" into a whole-class discussion- 10minutes **Session introduction (10 mins)** The facilitator will give brief presentation about the session **Interactive presentation: 80 mins** The facilitator will give an interactive presentation about **Assessing Recommendations** **Interactive presentation: 60 mins** Fascilitator and Barriers to the Implementation of Evidence into Practice	

Chapter Five: Assess Outcomes, Make Changes to Practice, and Quality Assurance

Duration - 4 Hrs

Enabling objectives: At the end of this session participants will be able to:

- Describe importance of assessing changes after application of evidence based practice
- Identify mechanisms of disseminating findings
- Perform self assement of changes or improvements using the self assement checklist
- Apply quality assurance strategies in EBM

5.1 Assess Outcomes and Make Changes to Practice

Session Duration: 2 ½ Hr

Specific Objectives: at the end of this session trainees are expected to assess Outcomes and make changes to daily clinical Practice

Summary of Activity: Assess Outcomes and Make Changes to Practice

No	Activity	Method of delivery	Time	Slide Number	Resource
3.	Introduction	• Interactive Lecture	20 min		LCD Flip Chart Computer markers
4.	How to measure changes	• Interactive Lecture • Think-Pair-Share	40 min		
5.	Disseminating the results of an EBP intervention	• Interactive Lecture	40 min		
6.	How to assess your own EBP performance	• Interactive Lecture	40 min		

Learning activities: Assess Outcomes and Make Changes to Practice

Time	Activity	Material and resources
60	The facilitator will introduce the session objectives and ask the trainees about the parameters used to assess changes after application of EBP? **Duration – 15min** **Instruction** o T: (Think) about what you know or have learned about the above question- 5minutes . o P: (Pair) with another participant or a small group and discuss points - 5minutes o S: (Share) your thinking with your partner and facilitators expand the "share" into a whole-class discussion- 10 minutes **Interactive lecture:** Introduction: 20 How to measure changes : 40 mins Disseminating the results of an EBP intervention 40 mins How to assess your own EBP performance:40	LCD and Computer Flip chart and marker Reference manual

Session 5.2 Quality Assurance

Duration - 90 min

Enabling objectives

- Describe quality indicators

Explain the steps of quality indicator assessment

Summary of Activity: quality indicators and assessment

No	Activity	Method of delivery	Time	Slide Number	Resource
7.	evidence-based medicine quality indicators and indicator assessment for identifying level of care	• Think Pair Share • Interactive Lecture	80 min		LCD Flip Chart Computer markers
8.	Summary	• Present the final session • Ask questions and comments.	10 min		

Learning activities: sources of evidence

	Topic	Time	Activity
1	Evidence-based medicine quality indicators and indicator assessment for identifying level of care	80 min	**Think-Pair-Share** Develop ideas for quality performance indicators to be used in your hospital/work area? **Duration – 20min** **Instruction** • T: (Think) about what you know or have learned about the above question- 3 minutes. • P: (Pair) with another participant or a small group and discuss points – 5 minutes. • S: (Share) your thinking with your partner and facilitators expand the "share" into a whole-class discussion- 7 minutes **Lecture 60 min** • Discuss quality indicators • Discuss quality indicator assessment
3.	Summary		• Present the final session • Ask questions and comments.

Annex 1: Think-Pair Share Template

Name _____ Date_____

Think-Pair-Share

Read the following question or problem:

Think

On your own, write three ideas you have about this question or problem:

1._____

2._____

3._____

Pair

Discuss your ideas with a partner. Put a check by any ideas, above, that your partner also wrote down. Then, write down ideas your partner had that you did not have:

1._____

2._____

3._____

Share

Review all of your ideas and circle the one you think is most important. One of you will share this idea with the whole group. As you listen to the ideas of the whole group, write down three more ideas you liked:

1._____

2._____

3._____

Annex 2: Gallery Walk

A gallery walk is a great way to engage participants in a training session. It can be used to introduce a new topic, review material, or even to assess understanding. Here are some instructions for conducting a successful gallery walk:

1. Prepare the space: Before the gallery walk, make sure the space is set up for success. Arrange the room so that participants can easily move around and view the materials. If possible, provide tables or stands to display the materials.

2. Prepare the materials: Gather the materials that you want to use for the gallery walk. These can include posters, handouts, pictures, videos, or any other visual aids. Make sure the materials are clearly labeled and organized in a way that makes sense.

3. Introduce the activity: Explain to the participants what a gallery walk is and how it will be conducted. Let them know that they will be moving around the room to view the materials and that they should take notes or ask questions as they go.

4. Divide into groups: Divide the participants into small groups and assign each group to a specific area of the room. Make sure each group has enough time to view the materials and discuss them.

5. Allow time for discussion: After each group has had a chance to view the materials, allow time for discussion. Encourage participants to ask questions and share their thoughts and ideas.

6. Wrap up: At the end of the gallery walk, review the materials and discuss any key points or takeaways. Make sure to thank the participants for their participation and encourage them to ask any remaining questions.

By following these instructions, you can ensure that your gallery walk is a successful and engaging activity for your training session. It can be a great way to introduce a new topic, review material, or assess understanding